

Geonu Lee

AI Research Engineer

✉ geonu.lee.dl@gmail.com  github.com/Geonu-Lee  linkedin.com/in/geonulee  geonu-lee.github.io

Work Experiences

Jul. 2024 - Present

Full-time – SNUAILAB

AI Research Engineer

- **Led the end-to-end development and deployment** of an AI-based smartphone cover glass micro-defect inspection system, achieving a False Negative Rate (FNR) below 2%.
- **Managed technical alignment and crisis resolution** with a manufacturing client, establishing technical trust that successfully secured strategic corporate investment.
- **Trained deep learning models** for a battery pole tracking system and integrated real-time pipelines into a C++ SDK, stabilizing edge deployment for a specialized inspection equipment manufacturer.
- **First-authored “Continual-MEGA”** (Preprint available on arXiv; currently under journal review).
- **Submitted a first-author paper on 3D anomaly detection** to ECCV 2026 (Under Rebuttal).

Oct. 2023 - Apr. 2024

Full-time – ALCHERA

AI Research Engineer, Anomaly Analysis Team

- **Developed and deployed DETR-based object detection models** for FireScout, a real-time vision-based wildfire detection solution, achieving a 20%p performance improvement over the legacy baseline.
- **Reconfigured the evaluation pipeline** from image-level to instance-level metrics, resolving legacy evaluation limitations to establish a precise and robust performance measurement system.

Mar. 2023 - Sep. 2023

Intern – Naver Cloud

AI Research Engineer, Image Vision Team

- **Contributed to the development of a face anti-spoofing service framework** for eKYC services, constructing a robust system capable of detecting spoofing attacks across diverse environmental conditions.
- **First-authored a paper published at WACV 2025** on object anti-spoofing, improving HTER by 7.5% and EER by 6.6% to significantly enhance domain generalization performance.

Jan. 2020 - Feb. 2020

Intern – Electronics and Telecommunications Research Institute (ETRI)

AI Research Engineer

- Developed a human action recognition system that models interactions between humans and surrounding objects in video data.

Research Interests

Deep Learning, Multi-Modal, Vision-Language-Model, Vision-Language-Action, Anomaly Detection, Domain Generalization, Face Anti-Spoofing, Pedestrian Attribute Recognition, Human Action Recognition, Human-Object Interaction Detection

Publications

- May. 2026 | **Geonu Lee**, YoungJoon Yoo, “[Title Anonymized] A Study on 3D Anomaly Detection,” in *European Conference on Computer Vision (ECCV)* (Under rebuttal)
- Feb. 2026 | **Geonu Lee**, Yujeong Oh, Geonhui Jang, Soyoung Lee, Jeonghyo Song, Sungmin Cha, YoungJoon Yoo, “Continual-MEGA: A Large-scale Benchmark for Generalizable Continual Anomaly Detection,” in *arXiv* (Under review)
- Feb. 2025 | **Geonu Lee**, Yonghyun Jeong, Haneol Jang, YoungJoon Yoo, “Domain-Generalized Object Anti-Spoofing: Bridging Gaps and Patch Selection for Robust Detection across Domains,” in *Winter Conference on Applications of Computer Vision (WACV)*
- Sep. 2022 | **Geonu Lee**, Kimin Yun, Jungchan Cho, “Occluded Pedestrian-Attribute Recognition for Video Sensors Using Group Sparsity,” in *Sensors*, vol. 22, no. 17, pp. 6626
- Aug. 2022 | **Geonu Lee** and Jungchan Cho, “STDP-Net: Improved Pedestrian Attribute Recognition Using Swin Transformer and Semantic Self-Attention,” in *IEEE ACCESS*, vol. 10, no. 1, pp. 82656 - 82667
- Feb. 2021 | **Geonu Lee**, Kimin Yun, and Jungchan Cho, “Improved Human-Object Interaction Detection through On-the-Fly Stacked Generalization,” in *IEEE ACCESS*, vol. 9, no. 1, pp. 34251-34263
- Feb. 2020 | Bhishan Bhandari, **Geonu Lee**, and Jungchan Cho, “Body-Part-Aware and Multitask-Aware Single-Image-Based Action Recognition,” in *Applied Science*, vol. 10, no. 4, pp. 1531-1548

Research Projects

- 2022 | **A Study on the Complex Human Attributes for Situation Understanding**
Master's Research Project – Gachon University
- Conducted research on recognizing interactive human attributes in video sequences under complex contexts.
 - Utilized Swin Transformer and Transformer decoder architectures for semantic reasoning.
 - Funded by Electronics and Telecommunications Research Institute (ETRI).
- 2021 | **A Study on the Understanding of Pedestrians**
Master's Research Project – Gachon University
- Developed a pedestrian attribute recognition method robust to occlusion.
 - Applied group sparsity regularization to handle various levels of visual obstruction.
 - Funded by Electronics and Telecommunications Research Institute (ETRI).
- 2020 | **A Study on the Understanding of Human-Object-Interactions**
Undergraduate Research Project – Gachon University
- Designed a novel deep neural architecture based on on-the-fly stacked generalization for HOI detection.
 - Focused on modeling dynamic interactions between humans and surrounding objects.
 - Funded by Electronics and Telecommunications Research Institute (ETRI).
- 2019 | **A Study on the Understanding of Human Situation Based on Deep Learning**
Undergraduate Research Project – Gachon University
- Proposed a multi-task learning framework combining human pose estimation and action recognition.
 - Addressed real-world situational understanding using contextual cues.
 - Funded by Electronics and Telecommunications Research Institute (ETRI).
- 2020 | **Development of Android Application for Dog Breed Prediction Using AI**
Undergraduate Project – Gachon University
- Developed a CNN-based dog breed classifier and deployed it in an Android application.
 - Implemented communication between Python back-end and Kotlin front-end using socket programming.

Education

Mar. 2021 - Feb. 2023	Gachon University <i>M.S. in Software Engineering</i>
Mar. 2016 - Feb. 2021	Gachon University <i>B.S. in Department of Computer Engineering</i>